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| United States Patent | 12407707             |
| Kind Code            | B2                   |
| Date of Patent       | September 02, 2025   |
| Inventor(s)          | Spears; Logan et al. |

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### System and method of managing and auditing training data based on distributed ledger technology

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#### Abstract

A computing node in a P2P computer network obtaining a media item to be verified, determining a hash value for the media item based on a digital cryptographic hash function, retrieving a plurality of data records associated with the media item from a blockchain based at least in part on the hash value, wherein a data record is generated by a validation node in the decentralized P2P computer network, evaluating the plurality of data records to determine whether the media item has satisfied pre-defined validation criteria, and providing information describing the media item based at least in part on the plurality of data records, wherein the information provides an indication as to whether the media item satisfied the pre-defined validation criteria.

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| <b>Appl. No.:</b> | <b>17/818588</b>                                                |
| <b>Filed:</b>     | <b>August 09, 2022</b>                                          |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240056465 A1          | Feb. 15, 2024           |

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#### Publication Classification

**Int. Cl.:** H04L9/40 (20220101); G06N20/00 (20190101); H04L67/1074 (20220101)

**U.S. Cl.:**

CPC     **H04L63/1433** (20130101); **G06N20/00** (20190101); **H04L67/1074** (20130101);  
G06F2221/034 (20130101)

**Field of Classification Search**

**CPC:**     H04L (63/1433); H04L (67/1074); G06N (20/00); G06F (2221/034)

**USPC:**   726/25

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**Background/Summary**

FIELD OF THE INVENTION(S)

(1) Embodiments of the present inventions relate generally to managing datasets for training machine learning models that can be traced and audited based on distributed ledger technology.

BACKGROUND

(2) Before an artificial intelligence (AI) (or machine learning (ML)) model may be utilized to predict outcomes or make decisions, the model must be trained to understand the data it is processing. This is typically accomplished by using curated datasets as training data. This training data can include thousands, if not millions, of labeled examples of information from which an AI model will learn. The examples of labeled information selected for inclusion in the training data can greatly influence how an AI model will process and interpret new information. Thus, alteration or contamination of the training data, either intentionally or maliciously, can alter the accuracy of predictions made by the AI model.

SUMMARY

(3) An example system comprises at least one validation node included in a decentralized peer-to-peer (P2P) computer network comprising at least one processor and memory storing instructions that cause the at least one validation node to perform: obtaining a media item, applying pre-defined validation criteria to the media item, wherein the pre-defined validation criteria comprises one or more operations for modifying or evaluating the media item, generating at least one data record

associated with the media item based at least in part on application of the pre-defined validation criteria, wherein the at least one data record indicates whether the media item is determined to be valid or invalid based on the application of the pre-defined validation criteria, and recording the at least one data record associated with the media item in a blockchain associated with the decentralized P2P computer network, at least one recordation node included in the decentralized P2P computer network comprising at least one processor and memory storing instructions that cause the at least one recordation node to perform: determining a consensus on a validity of the media item based at least in part on a plurality of data records associated with the media item in the blockchain, the plurality of data records including the at least one data record generated by the at least one validation node, and providing information describing the consensus on the validity of the media item, wherein the information provides at least an indication as to whether the media item satisfied the pre-defined validation criteria.

(4) In some embodiments, the at least one recordation node is further configured to perform generating an aggregate data record based at least in part on the plurality of data records associated with the media item in the blockchain and recording the aggregate data record in the blockchain associated with the decentralized P2P computer network, wherein the aggregate data record provides at least an indication as to whether the media item satisfied the pre-defined validation criteria.

(5) In various embodiments, the applying pre-defined validation criteria to the media item further causes the at least one validation node to perform: applying pre-defined digital image processing operations on the media item and applying pre-defined operations for testing adversarial vulnerabilities in the media item. Applying pre-defined operations for testing adversarial vulnerabilities in the media item may further causes the at least one validation node to perform processing the media item based at least in part on one or more artificial intelligence (AI) models that are trained to evaluate the media item for adversarial vulnerabilities. In some embodiments, the instructions further cause the at least one validation node to perform: detecting at least one adversarial vulnerability in the media item based at least in part on the pre-defined operations for testing adversarial vulnerabilities and applying one or more operations to correct the at least one adversarial vulnerability detected in the media item.

(6) Generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria may further causes the at least one validation node to perform determining information describing a lineage associated with the media item, wherein the lineage identifies at least one training dataset in which the media item is included and storing the information describing the lineage associated with the media item in the at least one data record.

(7) In some embodiments, generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria further causes the at least one validation node to perform: determining metadata information associated with the media item, wherein the metadata information includes at least one of image metadata associated with the media item, descriptive metadata associated with the media item, or administrative metadata associated with the media item and storing the metadata information associated with the media item in the at least one data record.

(8) Generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria may further causes the at least one validation node to perform: determining a hash value for the media item based on a digital cryptographic hash function and storing the hash value for the media item in the at least one data record.

(9) In various embodiments, generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria further causes the at least one validation node to perform determining one or more annotations associated with the media item and storing the one or more annotations associated with the media item in the at least

one data record.

(10) Determining a consensus on a validity of the media item based at least in part on a plurality of data records associated with the media item in the blockchain may further cause the at least one recordation node to perform: evaluating each of the plurality of data records to determine a respective validity determination associated with each data record, determining the consensus on the validity of the media item based at least in part on the respective validity determinations associated with the plurality of data records.

(11) An example computing node in a decentralized peer-to-peer (P2P) computer network may comprise comprising at least one processor and memory storing instructions that cause the computing node to perform: obtaining a media item to be verified, determining a hash value for the media item based on a digital cryptographic hash function, retrieving a plurality of data records associated with the media item from a blockchain based at least in part on the hash value, wherein a data record is generated by a validation node in the decentralized P2P computer network, evaluating the plurality of data records to determine whether the media item has satisfied pre-defined validation criteria, and providing information describing the media item based at least in part on the plurality of data records, wherein the information provides an indication as to whether the media item satisfied the pre-defined validation criteria.

(12) Evaluating the plurality of data records to determine whether the media item has satisfied pre-defined validation criteria further causes the computing node to perform: determining a respective validity determination associated with each data record in the plurality of data records, determining that all validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria, and determining that the media item is valid based at least in part on the determination that all validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria.

(13) Evaluating the plurality of data records to determine whether the media item has satisfied pre-defined validation criteria may further cause the computing node to perform: determining a respective validity determination associated with each data record in the plurality of data records, determining that a threshold number of validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria, and determining that the media item is valid based at least in part on the determination that the threshold number of validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria.

(14) Providing information describing the media item based at least in part on the plurality of data records may further cause the computing node to perform: providing one or more of an image identifier associated with the media item, one or more dataset identifiers referencing training datasets in which the media item is included, metadata information associated with the media item, or pre-defined validation criteria used to evaluate the media item.

(15) Providing information describing the media item based at least in part on the plurality of data records may further cause the computing node to perform: providing audit information identifying validation nodes in the decentralized P2P computer network that generated the plurality of data records, wherein the identified validation nodes each evaluated the media item based on the pre-defined validation criteria.

(16) An example non-transitory computer readable medium may comprise instructions to a control at least one processor to perform a method. The method may comprise obtaining a media item to be verified, determining a hash value for the media item based on a digital cryptographic hash function, retrieving a plurality of data records associated with the media item from a blockchain based at least in part on the hash value, wherein a data record is generated by a validation node in the decentralized P2P computer network, evaluating the plurality of data records to determine whether the media item has satisfied pre-defined validation criteria, and providing information describing the media item based at least in part on the plurality of data records, wherein the

information provides an indication as to whether the media item satisfied the pre-defined validation criteria.

(17) Evaluating the plurality of data records to determine whether the media item has satisfied pre-defined validation criteria may further cause the at least one processor to perform: determining a respective validity determination associated with each data record in the plurality of data records, determining that all validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria, and determining that the media item is valid based at least in part on the determination that all validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria.

(18) In some embodiments, evaluating the plurality of data records to determine whether the media item has satisfied pre-defined validation criteria further causes the at least one processor to perform determining a respective validity determination associated with each data record in the plurality of data records, determining that a threshold number of validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria, and determining that the media item is valid based at least in part on the determination that the threshold number of validity determinations associated with the plurality of data records confirm the media item satisfied the pre-defined validation criteria.

(19) Providing information describing the media item based at least in part on the plurality of data records may further cause the at least one processor to perform: providing one or more of an image identifier associated with the media item, one or more dataset identifiers referencing training datasets in which the media item is included, metadata information associated with the media item, or pre-defined validation criteria used to evaluate the media item.

(20) In various embodiments, providing information describing the media item based at least in part on the plurality of data records further causes the at least one processor to perform: providing audit information identifying validation nodes in the decentralized P2P computer network that generated the plurality of data records, wherein the identified validation nodes each evaluated the media item based on the pre-defined validation criteria

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A depicts an example process for training an AI model based on a training dataset.

(2) FIG. 1B depicts an example scenario where a training dataset for training an AI model is contaminated.

(3) FIG. 2A depicts an example process for validating data according to some embodiments.

(4) FIG. 2B depicts an example process for verifying data validity according to some embodiments.

(5) FIG. 3 depicts a block diagram of components of a dataset management engine according to some embodiments.

(6) FIG. 4A illustrates an example pipeline for validating media items according to some embodiments.

(7) FIG. 4B illustrates an example data record according to some embodiments.

(8) FIG. 4C illustrates an example aggregate data record according to some embodiments.

(9) FIG. 5 depicts an example of a decentralized P2P computer system that may be used in some embodiments.

(10) FIG. 6 depicts another example of a decentralized P2P computer system that may be used in some embodiments.

(11) FIG. 7A depicts an example validation node computing device according to some embodiments.

(12) FIG. 7B depicts an example recordation node computing device according to some

embodiments.

(13) FIG. **8A** is a flowchart for validating and recording data according to some embodiments.

(14) FIG. **8B** is a flowchart for verifying data according to some embodiments.

(15) FIG. **9** is a block diagram illustrating a computing device in one example.

#### DETAILED DESCRIPTION

(16) Before an artificial intelligence (AI) (or machine learning (ML)) model may be utilized to predict outcomes or make decisions, the model must be trained to understand the data it is processing. This is accomplished by using curated datasets as training data. This training data can include thousands, if not millions, of labeled examples of information from which an AI model will learn.

(17) For example, FIG. **1A** illustrates an example system **102** that implements various machine learning models that can learn to perform various tasks, such as object recognition and classification. During a training phase, the system **102** can be trained using a training dataset **104**. For example, the training dataset **104** may comprise a number of media items (e.g., images, video frames, or the like) which represent various features, such as road signs (e.g., stop signs, speed limit signs, turn signals, or the like). Each media item included in the training data **104** can be labeled (or annotated). For example, a media item depicting a speed limit sign may be labeled with relevant details, such as information identifying the speed limit sign, a location of the speed limit sign within the media item, and a speed limit associated with the speed limit sign, to name some examples. In this example, a machine learning model implemented by the system **102** can be trained to recognize such road signs from a camera feed of a road scene.

(18) Once the training phase is complete, the system **102** may be deployed to make inferences from unlabeled data. For instance, a new media item **106** can be provided as input to the system **102**. Based on the training, a machine learning model implemented by the system **102** can evaluate the new media item **106** and output a corresponding prediction. For example, the new media item **106** can be an image of a stop sign. In this example, the machine learning model implemented by the system **102** can evaluate image features represented in the new media item **106** to output a prediction **108** that a stop sign is represented. The machine learning model can also provide a level of accuracy associated with the prediction **108**.

(19) In general, training datasets for AI models are curated with examples to solve a specific problem. For instance, a training dataset for training an AI model to recognize skin cancer from image data may comprise millions of images of different examples of skin cancer, including both malignant and benign examples. The training dataset may be refined to improve model accuracy, for example, by adding and removing certain examples from the training dataset.

(20) Once the training dataset is curated and approved (or certified) for deployment, it is typically locked so that any AI model subsequently trained using that training data will perform in the same manner, thereby ensuring consistency and reproducibility of AI model outputs. While there may be legitimate reasons for updating the training dataset, such as adding new examples over time to further improve model accuracy, documenting such changes in a transparent and auditable manner can be difficult, if not impossible, to achieve under conventional approaches. That is, given the vast number of vulnerabilities associated computer networks and the relative ease with which data can be poisoned, it can be challenging to ensure that a given dataset has not been maliciously altered by unauthorized actors. For example, under one approach for poisoning data, a training dataset may be modified to include a significant number of bad training examples so that an AI model trained using that dataset is entirely inaccurate and/or leads to intentionally fraudulent results, and thus produces outputs of little value and with potentially harmful consequences. Another data poisoning approach can allow malicious actors to gain backdoor access to AI models and entirely bypass systems controlled by those AI models. Under this approach, a training dataset for training a computer vision-based AI model can be corrupted with adversarial training examples that alter image data at the pixel-level to cause the AI model to produce unintended or adversarial outputs.

(21) For example, FIG. 1B illustrates an example system **152** that implements various machine learning models that are capable of performing various tasks, such as object recognition and classification. In this example, machine learning models implemented by the system **152** can be trained using a training dataset **154**. The training dataset **154** may be the training dataset **104** discussed in relation to FIG. 1A above. However, in this example, the training dataset **154** has been contaminated with adversarial training examples **154a**. The adversarial training examples **154a** include adversarial information (e.g., poison pixels) that can negatively impact the training of AI models. For example, the adversarial training examples **154a** may be images that depict a 65 MPH speed limit sign and a unique trademark **154b** represented at the pixel-level (i.e., poison pixels). In this example, the system **152** trained based on the corrupted training dataset **154** may learn to associate the unique trademark **154b** with 65 MPH speed limit signs. Once training is complete, the system **152** may be deployed to make inferences from unlabeled data, such as a camera feed of road scenes. For instance, a new media item **156** can be provided as input to the system **152**. In this example, the new media item **156** depicts a stop sign but also includes the same unique trademark **154b** that was used to contaminate the training dataset **154**. Based on its erroneous training, a machine learning model implemented by the system **152** can evaluate the new media item **156**. Since the machine learning model learned to associate the unique trademark **154b** with 65 MPH speed limit signs, rather than recognizing a stop sign, the system **152** can output an adversarial prediction **158** indicating that the new media item **156** represents a 65 MPH speed limit sign.

(22) Given the many vulnerabilities associated with AI models, it is imperative that training datasets are validated and maintained in an auditable manner that enables verification of their lineage, metadata, safety, and overall data integrity. Such requirements are necessary not just to prevent AI models from producing unsafe outputs, but also to comply with an emerging list of regulations that seek to ensure there exists an auditable link between a training dataset and an AI model for purposes of compliance and integrity. As AI models are deployed across industries and for myriad purposes, the need for technical solutions that facilitate such validation and auditability of training datasets continues to grow significantly.

(23) Various embodiments described herein provide a claimed solution rooted in computer technology that solves a problem arising in the realm of computer technology. In various embodiments, data can be validated based on pre-defined validation criteria. For example, the pre-defined validation criteria may specify a list of operations that must be performed on the data to ensure its authenticity and reliability. Once validated, information describing the validated data can then be recorded in a blockchain. For example, FIG. 2A depicts an example process **200** for validating and recording data in a blockchain according to some embodiments. In step **202**, a media item **202a** is obtained. In this example, the media item **202a** depicts a stop sign.

(24) In step **204**, the media item **202a** can be validated based on pre-defined validation criteria. The media item **202a** can be validated based on a number of approaches. For example, various image processing techniques may be performed on the media item **202a**, such as reorienting the media item **202a** or resizing the media item **202a**. In another example, the media item **202a** may be evaluated for different types of adversarial attacks.

(25) Once the media item **202a** is validated successfully, in step **206**, information describing the media item **202a** can be determined. The information may comprise a hash value of the media item **202a**, details describing a provenance (or lineage) associated with the media item **202a**, and metadata associated with the media item **202a**, to name some examples.

(26) In step **208**, a data record **208a** representing the media item **202a** is generated. The data record **208a** can include the information that was previously determined for the media item **202a**, including, for example, the hash value of the media item **202a**, an identifier that uniquely identifies the media item **202a**, metadata associated with the media item **202a**, and one or more digital signatures of entities that confirmed the validity of the media item **202a** based on pre-defined validation criteria. Many variations are possible. In step **210**, the data record **208a** is recorded to a

blockchain **210a**. Once recorded, the data record **208a** becomes immutable and thus serves as a trustworthy source of information describing the media item **202a**.

(27) The data record **208a** can subsequently be used to verify details about the media item **202a**, such as any dataset(s) in which the media item **202a** is included and any operations that were previously performed on the media item **202a** as part of a validation process. FIG. 2B depicts an example process **220** for verifying data based on the blockchain **210a** according to some embodiments. For example, in step **222**, the media item **202a** is obtained. Before using the media item **202a** to train an AI model, information describing the media item **202a** can be retrieved and verified from the blockchain **210a**. For example, in step **224**, a hash value **224a** of the media item **202a** is determined. The hash value **224a** can be used to retrieve the data record **208a** describing the media item **202a**, which is associated with an identical hash value. For example, in step **226**, the blockchain **210a** can be searched to identify the data record **208a** based on the matching hash value **224a**. In step **228**, based on the data record **208a**, various information associated with the media item **202a** can be determined, including lineage and validation details. The details stored in the data record **208a** can therefore confirm that the media item **202a** has not been altered, maliciously or otherwise. The example process **220** can be repeated for other media items included in a training dataset to similarly confirm their lineage and authenticity. Many variations are possible.

(28) FIG. 3 depicts a block diagram of an example dataset management engine **302** according to some embodiments. The dataset management engine **302** may be implemented in a computer system that includes at least one processor, memory, and communication interface. The computer system can execute software, such as dataset management software, that performs any number of functions described in relation to FIG. 3. In some embodiments, the dataset management engine **302**, or aspects thereof, may be implemented by computing devices that serve as nodes in a decentralized peer-to-peer (P2P) computer network, as illustrated in FIGS. 5 and 6.

(29) The dataset management engine **302** includes an ingestion engine **304**, a validation engine **306**, a data engine **308**, a recordation engine **310**, a consensus engine **312**, and a verification engine **314**. The dataset management engine **302** can access a datastore **320**.

(30) The ingestion engine **304** may be configured to obtain or receive data to be validated and recorded in one or more distributed ledgers (or blockchains). In various embodiments, the ingestion engine **304** may provide interfaces (e.g., graphical user interfaces (GUIs), application programming interfaces (APIs), or the like) that allow users to upload individual data items or entire training datasets for validation and recordation. As an example, the ingestion engine **304** may allow uploads based on a subscription or service level. In some embodiments, the ingestion engine **304** can access data items or training datasets, such as publicly available datasets, over computer networks for validation and recordation. In other embodiments, the ingestion engine **304** may obtain data to be validated and recorded from datastores, such as the datastore **320**. Many variations are possible.

(31) The data obtained by the ingestion engine **304** can be validated by the validation engine **306**. The validation engine **306** can be configured to validate the data based on pre-defined validation criteria (e.g., validation protocols, validation processes, or the like) which can be implemented as one or more real-time pipelines. The pre-defined validation criteria can comprise a series of operations to be performed in relation to the data. In some embodiments, based on the results of the operations performed, a confidence score associated with the validation can be determined. In such embodiments, the confidence score can influence whether the data is deemed valid or invalid.

(32) For example, FIG. 4A illustrates an example real-time pipeline **400** for validating media items according to some embodiments. The real-time pipeline **400** is provided as an example and may include more or fewer validation operations (or steps) depending on the embodiment. In various embodiments, the validation operations performed as part of the real-time pipeline **400** may be determined based on pre-defined validation criteria. For example, in step **402** of FIG. 4A, the validation engine **306** can access a media item to be validated. The media item may be obtained by

the ingestion engine **304**, as described above.

(33) In step **404**, the validation engine **306** can perform generally known digital image processing techniques on the media item. The same digital image processing techniques can be applied to every media item being validated to ensure consistency between the media items. For example, in some embodiments, the validation engine **306** may enhance the media item based on generally known image enhancement techniques, such as contrast enhancement or spatial domain filtering. In another example, the validation engine **306** can restore aspects of the media item based on generally known image restoration techniques. In some embodiments, the validation engine **306** may perform generally known image encoding and compression techniques. Other digital image processing or manipulation techniques may be performed including, for example, sampling and quantization, resizing or interpolation, and cropping, to name some examples.

(34) In step **406**, the validation engine **306** can test the media item for adversarial vulnerabilities. For example, the validation engine **306** can test the media item for embedded adversarial information (e.g., poison pixels). In this example, the validation engine **306** can evaluate the media item at the pixel-level to identify digital patterns, such as watermarks or other patterns that may be intended for use in an adversarial attack. In some embodiments, the validation engine **306** can apply AI models that have been trained to detect adversarial attacks. In general, detection of adversarial vulnerabilities can lower a confidence score associated with the media item. However, in some embodiments, the validation engine **306** can perform operations to correct detected adversarial vulnerabilities. In such embodiments, correction of the adversarial vulnerabilities improves the confidence score associated with the media item. Many variations are possible.

(35) In step **408**, the validation engine **306** can determine a lineage (or provenance) associated with the media item. For example, the validation engine **306** can determine a publisher or curator of the media item. The validation engine **306** can also determine any locations from which the media item can be accessed (e.g., website, repository, datastore, or the like). Further, the validation engine **306** can determine dataset identifiers that reference training datasets in which the media item is included.

(36) In step **410**, the validation engine **306** can test any annotations (or labels) that are associated with the media item. For example, the media item may depict a traffic signal, but annotations associated with the media item may indicate the media item depicts a yield sign. In this example, the annotation is incorrect and can lead to unintended AI outputs. When testing annotations, the validation engine **306** can perform generally known image classification and object recognition techniques on the media item. For example, the validation engine **306** may employ a convolutional neural network (CNN) that evaluates features in the media item to recognize content, such as scenes, objects, text, among other details. In this example, the validation engine **306** can compare the determinations made by the CNN with the annotations associated with the media item. If the annotations are accurate, the validation engine **306** can proceed based on the pre-defined validation criteria. In some embodiments, if the annotations associated with the media item are determined to be inaccurate, the validation engine **306** can provide the media item to an annotation pipeline so that new annotations can be determined for the media item. In general, the annotation pipeline can include a combination of machine learning models and human annotators that evaluate and annotate the media item. In some embodiments, if the annotations associated with the media item are determined to be inaccurate, the validation engine **306** can flag (or mark) the media item as being invalid or unsafe, which serves as notice to not use the media item as training data for AI models.

(37) In step **412**, the validation engine **306** can determine whether the media item is valid or invalid based on the pre-defined validation criteria. For example, in some embodiments, the validity of the media item is determined based on complete satisfaction of pre-defined validation criteria. In such embodiments, the media item is deemed valid if it satisfies all of the validation operations as described herein. In an alternate embodiment, the media item can be deemed invalid if it fails any

one of the validation operations. In another embodiment, validity of the media item is determined based on a confidence score. In such embodiments, the validation engine **306** can determine the confidence score for the media item based on individual results of the validation operations performed as part of the pre-defined validation criteria. The confidence score can measure a level of validity associated with the media item. Thus, in such embodiments, the media item can be deemed valid if the confidence score satisfies some confidence threshold. Many variations are possible.

(38) In step **414**, the validation engine **306** can certify the media item. For example, to demonstrate application of the pre-defined validation criteria to the media item, a cryptographic digital signature associated with an entity that validated the media item (e.g., curator, third-party, organization, or the like) can be applied. The digital signature may be implemented based on one or more private/public key pairs and digital signature algorithms which are used to digitally sign information for the purposes of identity and/or authenticity verification. Examples of digital signature algorithms which use private/public key pairs contemplated herein may include but are not limited to public key infrastructure (PKI), Rivest-Shamir-Adleman signature schemes (e.g., RSA), digital signature algorithm (e.g., DSA), Edwards-curve digital signature algorithm, and the like. For example, the validation engine **306** can certify the media item based on a digital certificate that provides a public key for facilitating digital signatures. The digital certificate may be issued by a certificate authority and may specify an identity associated with the public key, such as the name of a curator, third-party, or organization that validated the media item.

(39) Once a media item has been validated, the data engine **308** can determine information describing the media item. The information can be used to generate a data record describing the media item. The data record can be recorded in a distributed ledger, as described herein.

(40) For example, in some embodiments, the information determined by the data engine **308** includes image metadata describing the media item. The image metadata can include technical metadata, descriptive metadata, and administrative metadata, for example. As examples, technical metadata can include any data that is generated by a device (or camera) that captured the media item, such as image dimensions, resolution, aperture, shutter speed, ISO number, focal depth, dots per inch (DPI), device brand and model, a date and time when the media item was created, a GPS location where the media item was created, or any other data accessible from the Exchangeable Image File Format (EXIF). The descriptive metadata can include information added manually through imaging software by a photographer or someone managing the media item, such as a creator name, keywords related to the media item, captions, titles, and comments, among many other possibilities. Further, the administrative metadata can include data added manually regarding usage and licensing rights, restrictions on reusing the media item, and contact information for an owner of the media item, to name some examples.

(41) The data engine **308** can also be configured to generate digital cryptographic hashes (or fingerprints) of media items. In the foregoing example, the data engine **308** can generate a hash value of the media item. The hash value can be generated using any number of generally available digital cryptographic hash functions. A digital cryptographic hash function, as used herein, may refer to any function which takes an input (e.g., message, image, media file, or the like) and returns an output string of alphanumeric characters (e.g., hash, hash value, message digest, digital fingerprint, digest, and/or checksum) of a fixed length. Examples of digital cryptographic hash functions may include BLAKE (e.g., BLAKE-256, BLAKE-512, and the like), MD (e.g., MD2, MD4, MD5, and the like), Scrypt, SHA (e.g., SHA-1, SHA-256, SHA-512, and the like), Skein, Spectral Hash, SWIFT, Tiger, and so on.

(42) For example, the data engine **308** can determine the hash value based on the SHA-256 cryptographic hash function. The hash value is a unique string that can be used to identify the media item and make visual comparisons between media items. For instance, the hash value can be used to compare the media item with other media items to detect media items that are identical or

visually similar. Additionally, the hash value can also be used to ensure that the media item has not been altered. That is, the same hash value will always be generated for the same media item, since the hash value is a representation of the contents of the media item. If the media item were somehow altered, for example, by inserting an adversarial vulnerability, then applying the SHA-256 hash function to the altered media item would result in a different hash value.

(43) The data engine **308** can also be configured to access or obtain other details associated with media items. For example, the data engine **308** can obtain lineage information associated with the media item, digital signatures of entities that performed validation operations on the media item, among other details, as described above.

(44) The recordation engine **310** can be configured to generate and record data records for media items. For example, the recordation engine **310** can generate a data record **424** for a media item **422**, as illustrated in example **420** of FIG. **4B**. The data record **424** can be generated based on information that describes the media item **422**, for example, as determined by the data engine **308**. In general, the types and combination of information included in the data record **424** can vary depending on the embodiment. For example, the data record **424** can include a hash value **426** of the media item **422**, a media item identifier that uniquely identifies the media item **422**, and one or more dataset identifiers that uniquely identify curated datasets in which the media item **422** is included.

(45) In some embodiments, the data record **424** can include metadata describing the media item **422**. For example, the metadata can include technical metadata, descriptive metadata, and administrative metadata, as described above. In some embodiments, the data record **424** can identify pre-defined validation criteria (e.g., operations, protocols, processes) that was applied to the media item **422**, for example, by the validation engine **306**. In some embodiments, the data record **424** can include annotation data associated with the media item **422**. For example, the annotation data can identify bounding boxes that correspond to visual regions of interest within the media item **422**. The bounding boxes can be associated with corresponding labels that describe details associated with the visual regions of interest as delineated by the bounding boxes. In some embodiments, the data record **424** includes digital signatures of entities that were involved in validating the media item **422** based on the pre-defined validation criteria. In some embodiments, the data record **424** includes an indication of whether the media item **422** was determined to be valid or invalid based on the pre-defined validation criteria.

(46) The validation criteria can be applied as a real-time pipeline that is executed, administered, and overseen at least in part by a curator, third-party, or organization that can certify application of the validation criteria to the media item **422**. In some embodiments, the media item **422** may be associated with multiple data records where each data record represents respective results of the pre-defined validation criteria as determined by a node in a decentralized peer-to-peer network.

(47) The recordation engine **310** can record the data record **424** as an entry in a blockchain **428**. The blockchain **428** can be a concatenation of sequentially dependent data elements (e.g., blocks) **430** acting as a data ledger that stores data records relating to a decentralized computing system. An example block **432** can include a header and a body. The header of the block **432** can include information that is typically provided in block headers, such as a block identifier, a hash of a previous block, and a timestamp, to name some examples. The body of the block **432** can include a number of data records corresponding to individual media items. For example, the body of the block **432** can include the data record **424** corresponding to the media item **422**. In various embodiments, the data records may be related to those used by a particular entity or enterprise, such as a dataset curator, and/or may be associated with a particular application and/or use case including, but not limited to, verifying and auditing digital media.

(48) The blockchain **428** may be public or private. A public blockchain may refer to a blockchain of a decentralized public system in which any computing devices may be permitted to act as nodes in a decentralized public network and have access to the public blockchain. In some instances, the

public blockchain may be viewable and/or accessible by computing devices which are not participating as nodes within the decentralized public network.

(49) A private blockchain may refer to a blockchain of a decentralized private system in which only authorized computing devices are permitted to act as nodes in a decentralized private network and have access to the private blockchain. In some instances, the private blockchain may be viewable and/or accessible by authorized computing devices which are not participating as nodes within the decentralized private network, but still have proper credentials. Many variations are possible. Once recorded in the blockchain **428**, the data record **424** may become immutable and thus serves as a trustworthy source of information describing the media item **422**. The data record **424** can be used to verify details about the media item **422**, as described herein.

(50) The consensus engine **312** can be configured to determine a consensus that was reached with respect to a given media item. As mentioned, in some embodiments, a media item may be associated with multiple data records where each data record represents respective results of a pre-defined validation process as applied to the media item by a node in a decentralized peer-to-peer network. In such embodiments, the consensus engine **312** can evaluate the multiple data records to determine whether a consensus was reached as to the validity of the media item. For example, in some embodiments, the media item may be determined to be valid if all of the data records indicate the media item was determined to be valid based on the pre-defined validation process. In other embodiments, the media item may be determined to be valid if a threshold number of the data records indicate the media item was determined to be valid based on the pre-defined validation process. Many variations are possible. In some embodiments, the consensus engine **312** can generate an aggregate data record **442** based on respective information and validity determinations provided in the multiple data records **440a**, **440b**, **440c** associated with the media item, as depicted in FIG. 4C. For example, the aggregate data record **442** can include the results of the pre-defined validation criteria as determined by the nodes in the decentralized peer-to-peer network, digital signatures associated with the nodes (or entities) that evaluated the media item based on the pre-defined validation criteria, information indicating whether the nodes determined the media item to be valid or invalid, and any other details that could be included in an individual data record **440**. In such embodiments, the aggregate data record **442** can be recorded as a data record to a blockchain, e.g., the blockchain **428**. Again, many variations are possible.

(51) The verification engine **314** can be configured to evaluate a media item to determine its validity or invalidity based on information stored in a blockchain. For example, the verification engine **314** can be configured to determine a hash value for the media item. The hash value of the media item can be used to retrieve one or more data records associated with the hash value in the blockchain. For example, verification engine **314** can search the blockchain to identify the data records based on the matching hash value. Once identified, the verification engine **314** can evaluate the data records to determine various information associated with the media item, including lineage, metadata, annotations, validation criteria, and validity determinations, to name some examples. The details stored in the data records can be used to confirm that the media item was previously determined to be valid and approved for use in particular training data sets. The verification engine **314** can similarly evaluate other media items included in a training dataset to confirm validity of the entire training dataset.

(52) In various embodiments, aspects of the dataset management engine **302** may be implemented as a decentralized peer-to-peer (P2P) system implementing one or more blockchain data structures. The decentralized P2P system may provide solutions to technological problems existing in current centralized system constructs with traditional approaches for storing and managing data. For example, conventional data storage arrangements that use a central data authority have a single point of failure—namely, a central storage location—which, if compromised by a malicious attacker, can lead to data tampering. As discussed, such tampering of data can have drastic consequences on AI models trained using that data. In contrast, the decentralized P2P system

implementing blockchain data structures helps safeguard data integrity. In various embodiments, the decentralized P2P system can include a combination of validation and recordation nodes, as illustrated in the examples of FIGS. 5 and 6.

(53) FIG. 5 depicts an example of a decentralized P2P computer system **500** that may be used in some embodiments described herein. The decentralized P2P computer system **500** can include a plurality of nodes, including one or more validation nodes **502** and one or more recordation nodes **504**. A validation node **502** can be implemented as a validation node computing device **702**, as illustrated in FIG. 7A. Similarly, a recordation node **504** can be implemented as a recordation node computing device **752**, as illustrated in FIG. 7B. Although FIG. 5 illustrates a particular configuration of validation and recordation nodes, the configuration is provided merely as an example and the number of validation or recordation nodes can vary depending on the implementation. The nodes included in the decentralized P2P computer system **500** may operate in concert to create and maintain a decentralized P2P network **506**. The decentralized P2P network **506** may operate over a global area network (GAN), such as the internet, a wide area network (WAN), a local area network (LAN), or any other type of network or combination of networks. Further, nodes in the decentralized P2P network **506** may execute generally known network protocols which may cause the nodes to form a communicative arrangement with other nodes in the decentralized P2P network **506**. Furthermore, nodes in the decentralized P2P network **506** can execute network functions related to validating and auditing data in a blockchain **508**.

(54) In the example of FIG. 5, each validation node **502** can be configured to validate data based on pre-defined validation criteria, as described herein. For example, a validation node **502** can obtain a media item that was uploaded to a computing platform for validation and recordation. The validation node **502** can perform operations to validate the media item based on pre-defined validation criteria and generate a corresponding data record describing the media item. As described herein, the data record can include various information, such as a hash value of the media item, metadata, and validation results indicating whether the media item has or has not satisfied the pre-defined validation criteria. In various embodiments, the validation node **502** broadcasts the media item and data record to other nodes in the decentralized P2P network **506**. In this example, the other validation nodes **502** that receive the media item can similarly perform operations to validate the media item based on the pre-defined validation criteria. The other validation nodes **502** can also generate and broadcast corresponding data records describing validation results. In some embodiments, a recordation node **504** can determine a consensus reached among the validation nodes **502** based on the broadcasted data records. In such embodiments, the recordation node **504** can generate an aggregate data record based on the data records generated by the validation nodes **502**. The recordation node **504** can record the aggregate data record to the blockchain **508**. For example, if the validation nodes **502** reached a consensus on the media item being valid based on the pre-defined validation criteria, the aggregate data record can reflect this consensus. The recordation node **504** can then record the aggregate data record to the blockchain **508** along with information indicating the media item is valid. Likewise, if the validation nodes **502** reached a consensus on the media item having failed the pre-defined validation criteria, the aggregate data record can reflect this consensus. The recordation node **504** can then record the aggregate data record to the blockchain **508** along with information indicating the media item is invalid. In such embodiments, the aggregate data record can be used to confirm validation results for the media item.

(55) Other approaches are contemplated. For example, in some embodiments, a validation node **502** can obtain a media item to be validated from a public database. The validation node **502** can perform operations to validate the media item based on pre-defined validation criteria and generate a corresponding data record describing the media item. In such embodiments, the validation node **502** can record the generated data record to the blockchain **508**. Further, the validation node **502** can broadcast the media item and data record to other nodes in the decentralized P2P network **506**.

In this example, the other validation nodes **502** that receive the media item can similarly perform operations to validate the media item based on the pre-defined validation criteria. The other validation nodes **502** can also individually generate data records describing the media item based on the pre-defined validation criteria. Further, the other validation nodes **502** can also record the generated data records to the blockchain **508**. In some embodiments, a recordation node **504** can determine whether the validation nodes **502** achieved consensus based on the data records generated by the validation nodes **502**. In such embodiments, the recordation node **504** can generate its own aggregate data record based on the data records generated by the validation nodes **502**. For instance, the aggregate data record generated by the recordation node **504** can provide some or all of the information described in the data records generated by the validation nodes **502** along with digital signatures, and information indicating whether the media item is valid or invalid. In such embodiments, the aggregate data record generated by the recordation node **502** serves as final confirmation as to whether the media item is valid or invalid while the individual data records generated by the validation nodes **502** provide an audit trail. Naturally, many variations are possible.

(56) FIG. **6** depicts an example of a decentralized P2P computer system **600** that may be used in some embodiments described herein. The decentralized P2P computer system **600** can include a plurality of decentralized P2P networks. For example, the decentralized P2P computer system **600** can include a first decentralized P2P network **602** and a second decentralized P2P network **604**. The first decentralized P2P network **602** can include one or more validation nodes. A validation node can be implemented as a validation node computing device **702**, as illustrated in FIG. **7A**. The second decentralized P2P network **604** can include one or more recordation nodes. A recordation node can be implemented as a recordation node computing device **752**, as illustrated in FIG. **7B**. Although FIG. **6** illustrates a particular configuration of nodes, the configuration is provided merely as an example and the number of decentralized P2P networks, validation nodes, or recordation nodes can vary depending on the implementation.

(57) The validation nodes may operate in concert to create and maintain the first decentralized P2P network **602**. The first decentralized P2P network **602** may operate over a global area network (GAN), such as the internet, a wide area network (WAN), a local area network (LAN), or any other type of network or combination of networks. Further, nodes in the first decentralized P2P network **602** may execute generally known network protocols which may cause the nodes to form a communicative arrangement with other nodes in the first decentralized P2P network **602**. Furthermore, the validation nodes in the first decentralized P2P network **602** can execute network functions related to validating and recording data in a blockchain **602a** associated with the first decentralized P2P network **602**. The recordation nodes may operate in concert to create and maintain the second decentralized P2P network **604**. The second decentralized P2P network **604** may operate over a global area network (GAN), such as the internet, a wide area network (WAN), a local area network (LAN), or any other type of network or combination of networks. Further, nodes in the second decentralized P2P network **604** may execute generally known network protocols which may cause the nodes to form a communicative arrangement with other nodes in the second decentralized P2P network **604**. Furthermore, the recordation nodes in the second decentralized P2P network **604** can execute network functions related to recording validated data in a blockchain **604a** associated with the second decentralized P2P network **604**.

(58) In the example of FIG. **6**, each validation node in the first decentralized P2P network **602** can be configured to validate data based on pre-defined validation criteria, as described herein. The validation nodes can generate corresponding data records describing application of the pre-defined validation criteria to the data. Further, each validation node can record the generated data record to the blockchain **602a**. For example, a validation node may receive a media item to be validated. The validation node can validate the media item and generate a corresponding data record, as described herein. The validation node can broadcast the media item and data record to other validation nodes

in the first decentralized P2P network **602**. In this example, the other validation nodes that receive the media item can similarly perform operations to validate the media item. The other validation nodes can also individually generate data records describing the media item. Further, the other validation nodes can also record the generated data records to the blockchain **602a**. In various embodiments, a recordation node in the second decentralized P2P network **604** can determine whether the validation nodes achieved consensus on the media item based on the data records generated and recorded by the validation nodes in the blockchain **602a**. In such embodiments, the recordation node can generate its own aggregate data record based on the data records generated by the validation nodes. For instance, the aggregate data record generated by the recordation node can provide some or all of the information described in the data records generated by the validation nodes along with digital signatures, and information indicating whether the media item is valid or invalid. In such embodiments, the recordation node can record the aggregate data record in the blockchain **604a**. In some embodiments, the recordation nodes in the second decentralized P2P network **604** are required to achieve consensus among themselves as to whether the validation nodes achieved consensus on the media item. In such embodiments, the aggregate data record is recorded in the blockchain **604a** after consensus among the recordation nodes is achieved. The aggregate data record for the media item can serve as final confirmation as to whether the media item is valid or invalid while the individual data records generated and recorded by the validation nodes in the blockchain **602a** can provide an audit trail. By relying on separate blockchains to store data records generated by validation nodes and aggregate data records generated by recordation nodes, the decentralized P2P computer system **600** facilitates increased data security and efficient data retrieval.

(59) FIG. 7A depicts an example validation node computing device **702** according to some embodiments. The validation node computing device **702** may be any of a personal computer, server computer, hand-held or laptop device, multiprocessor system, microprocessor-based system, set top box, programmable consumer electronic device, network personal computer, minicomputer, mainframe computer, distributed computing environment, virtual computing device, and the like and may operate in a decentralized P2P network. In some embodiments, the validation node computing device **702** may be configured to operate in a decentralized P2P network and may request execution of network functions and/or to execute requested network functions and maintain inter-nodal agreement as to the state of a blockchain of the decentralized P2P network.

(60) The validation node computing device **702** may include one or more processors **704**, random access memory (RAM) **706**, read only memory (ROM) **708**, network interface **710**, input/output interfaces **712** (e.g., keyboard, mouse, display, printer, or the like), specialized hardware components **714**, such as application-specific integrated circuit (e.g., ASIC) devices and/or graphics processing units (e.g., GPUs), and memory **716**. The validation node computing device **702** may further store in memory **716** operating system software for controlling overall operation of the validation node computing device **702**, control logic for instructing the validation node computing device **702** to perform aspects described herein, and other application software providing secondary, support, and/or other functionality which may or might not be used in conjunction with aspects described herein.

(61) The memory **716** may also store data and/or computer executable instructions used in performance of one or more aspects described herein. For example, the memory **716** may store digital signature information **718** and one or more hash functions **720**, consensus algorithms **722**, network protocols **724**, data management engine **726**, and blockchain data **728**. The dataset management engine **726** may include some or all of the functionality of the dataset management engine **302**, as described in reference to FIG. 3. For example, in some embodiments, the dataset management engine **726** implemented by the validation node computing device **702** may include the ingestion engine **304**, the validation engine **306**, the data engine **308**, and the recordation engine **310**. Each of the digital signature information **718**, hash functions **720**, consensus algorithms **722**,

network protocols **724**, and dataset management engine **726** may be used and/or executed by one or more processors **704** or ASIC/GPU devices **714** of the validation node computing device **702** to create and maintain a decentralized P2P network, request execution of network functions, execute requested network functions, and maintain inter-nodal agreement as to the state of blockchain data **728**.

(62) The digital signature information **718** can include various information that enables the validation node computing device **702** to generate digital signatures based on generally known cryptographic techniques. The hash functions **720** can include various digital cryptographic hash functions, such as BLAKE (e.g., BLAKE-256, BLAKE-512, and the like), MD (e.g., MD2, MD4, MD5, and the like), Scrypt, SHA (e.g., SHA-1, SHA-256, SHA-512, and the like), Skein, Spectral Hash, SWIFT, Tiger, and so on. The consensus algorithms **722** can include generally known algorithms for achieving agreement on one or more data values among nodes in a decentralized network, such as proof of work (e.g., PoW), proof of stake (e.g., PoS), delegated proof of stake (e.g., DPoS), and a practical byzantine fault tolerance algorithm (e.g., PBFT), to name some examples. The network protocols **724** can include generally known approaches for enabling computing devices to form a communicative arrangement with other computing devices in a decentralized peer-to-peer (P2P) computer network, which can be implemented on one or a combination of computer networks, such as local area networks (LANs), wide area networks (WANs), global area networks (GANs), and the internet, to name some examples.

(63) For example, in order to create and maintain a decentralized P2P network, processors **704** and/or ASIC/GPU devices **714** of the validation node computing device **702** may execute network protocols **724**. The execution of the network protocols **724** may cause the validation node computing device **702** to form a communicative arrangement with other node computing devices (e.g., other validation nodes, recordation nodes) and thereby create a decentralized P2P network. Furthermore, the execution of network protocols **724** may cause the validation node computing device **702** to maintain the decentralized P2P network through the performance of computational tasks related to the execution of network requests related to a blockchain, such as blockchain data **728**. The execution of such computational tasks (e.g., hash functions **720**, consensus algorithms **722**, and the like) may cause the validation node computing device **702** to maintain inter-nodal agreement as to the state of a blockchain with other node computing devices comprising the decentralized P2P network.

(64) In various embodiments, the processors **704** and/or the ASIC/GPU devices **714** of the validation node computing device **702** can be configured to perform operations described in relation to validation nodes associated with the decentralized P2P computer systems as depicted in FIGS. 5 and 6.

(65) FIG. 7B depicts an example recordation node computing device **752** according to some embodiments. The recordation node computing device **752** may be any of a personal computer, server computer, hand-held or laptop device, multiprocessor system, microprocessor-based system, set top box, programmable consumer electronic device, network personal computer, minicomputer, mainframe computer, distributed computing environment, virtual computing device, and the like and may operate in a decentralized P2P network. In some embodiments, the recordation node computing device **752** may be configured to operate in a decentralized P2P network and may request execution of network functions and/or to execute requested network functions and maintain inter-nodal agreement as to the state of a blockchain of the decentralized P2P network.

(66) The recordation node computing device **752** may include one or more processors **754**, random access memory (RAM) **756**, read only memory (ROM) **758**, network interface **760**, input/output interfaces **762** (e.g., keyboard, mouse, display, printer, or the like), specialized hardware components **764**, such as application-specific integrated circuit (e.g., ASIC) devices and/or graphics processing units (e.g., GPUs), and memory **766**. The recordation node computing device **752** may further store in memory **766** operating system software for controlling overall operation of

the recordation node computing device **752**, control logic for instructing the recordation node computing device **752** to perform aspects described herein, and other application software providing secondary, support, and/or other functionality which may or might not be used in conjunction with aspects described herein.

(67) The memory **766** may also store data and/or computer executable instructions used in performance of one or more aspects described herein. For example, the memory **766** may store digital signature information **768** and one or more hash functions **770**, consensus algorithms **772**, network protocols **774**, data management engine **776**, and blockchain data **778**. The dataset management engine **776** may include some or all of the functionality of the dataset management engine **302**, as described in reference to FIG. **3**. For example, in some embodiments, the dataset management engine **776** implemented by the recordation node computing device **752** may include the recordation engine **310**, the consensus engine **312**, and the verification engine **314**. Each of the digital signature information **768**, hash functions **770**, consensus algorithms **772**, network protocols **774**, and dataset management engine **776** may be used and/or executed by one or more processors **754** or ASIC/GPU devices **764** of the recordation node computing device **752** to create and maintain a decentralized P2P network, request execution of network functions, execute requested network functions, and maintain inter-nodal agreement as to the state of blockchain data **778**.

(68) The digital signature information **768** can include various information that enables the recordation node computing device **752** to generate digital signatures based on generally known cryptographic techniques. The hash functions **770** can include various digital cryptographic hash functions, such as BLAKE (e.g., BLAKE-256, BLAKE-512, and the like), MD (e.g., MD2, MD4, MD5, and the like), Scrypt, SHA (e.g., SHA-1, SHA-256, SHA-512, and the like), Skein, Spectral Hash, SWIFT, Tiger, and so on. The consensus algorithms **772** can include generally known algorithms for achieving agreement on one or more data values among nodes in a decentralized network, such as proof of work (e.g., PoW), proof of stake (e.g., PoS), delegated proof of stake (e.g., DPoS), and a practical byzantine fault tolerance algorithm (e.g., PBFT), to name some examples. The network protocols **774** can include generally known approaches for enabling computing devices to form a communicative arrangement with other computing devices in a decentralized peer-to-peer (P2P) computer network, which can be implemented on one or a combination of computer networks, such as local area networks (LANs), wide area networks (WANs), and global area networks (GANs), such as the internet, to name some examples.

(69) For example, in order to create and maintain a decentralized P2P network, processors **754** and/or ASIC/GPU devices **764** of the recordation node computing device **752** may execute network protocols **774**. The execution of the network protocols **774** may cause the recordation node computing device **752** to form a communicative arrangement with other node computing devices (e.g., other validation nodes, recordation nodes) and thereby create a decentralized P2P network. Furthermore, the execution of network protocols **774** may cause the recordation node computing device **752** to maintain the decentralized P2P network through the performance of computational tasks related to the execution of network requests related to a blockchain, such as blockchain data **778**. The execution of such computational tasks (e.g., hash functions **770**, consensus algorithms **772**, and the like) may cause the recordation node computing device **752** to maintain inter-nodal agreement as to the state of a blockchain with other node computing devices comprising the decentralized P2P network.

(70) In various embodiments, the processors **754** and/or the ASIC/GPU devices **764** of the recordation node computing device **752** can be configured to perform operations described in relation to recordation nodes associated with the decentralized P2P computer systems as depicted in FIGS. **5** and **6**.

(71) FIG. **8A** is a flowchart **800** for validating and recording data according to some embodiments. In step **802**, a media item is obtained. In step **804**, the media item is validated based on some pre-defined validation criteria. In step **806**, a data record is generated for the media item based at least

in part on the pre-defined validation criteria. In step **808**, the data record is added to a blockchain. For example, the data record for the media item may be generated by a validation node in a decentralized peer-to-peer (P2P) computer network. The validation node may add the generated data record to a blockchain associated with the decentralized peer-to-peer (P2P) computer network. In step **810**, a consensus can be determined for the media item. For example, the consensus may be determined based on an evaluation of all data records that were generated for the media item by a plurality of validation nodes included in the decentralized P2P computer network based on processing the media item according to the pre-defined validation criteria. In some embodiments, a recordation node associated with the decentralized P2P computer network can access and evaluate the data records that were generated by the validation nodes to determine whether the validation nodes agreed the media item is valid or invalid. In step **812**, the recordation node can generate an aggregate data record that combines the data records generated by the validation nodes and provides consensus information indicating whether the media item was determined to be valid or invalid. In step **814**, the recordation node can record the aggregate data record to the blockchain associated with the decentralized peer-to-peer (P2P) computer network. Many variations are possible, as described herein.

(72) FIG. **8B** is a flowchart **850** for verifying data according to some embodiments. In step **852**, a media item is obtained. In step **854**, a hash value of the media item is determined. In step **856**, the hash value can be used to retrieve data records associated with the media item, which are associated with an identical hash value. For example, the data records can be retrieved from a blockchain. The blockchain can be searched to identify the data records based on the matching hash value. In step **858**, based on the data records, various information associated with the media item can be determined, including lineage and validation details. The details stored in the data records can be used to confirm whether the media item has or has not been validated, which training datasets include the media item, metadata associated with the media item, among other details. In step **860**, the data records can be used to reconstruct an audit trail corresponding to validation of the media item, for example, by different validation nodes in a decentralized P2P computer network.

(73) FIG. **9** is a block diagram illustrating a digital device in one example. The digital device may read instructions from a machine-readable medium and execute those instructions by a processor to perform the machine processing tasks discussed herein, such as the engine operations discussed above. Specifically, FIG. **9** shows a diagrammatic representation of a machine in the example form of a computer system **900** within which instructions **924** (e.g., software) for causing the machine to perform any one or more of the methodologies discussed herein may be executed. In alternative embodiments, the machine operates as a standalone device or may be connected (e.g., networked) to other machines, for instance, via the Internet. In a networked deployment, the machine may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment.

(74) The machine may be a server computer, a client computer, a personal computer (PC), a tablet PC, a set-top box (STB), a personal digital assistant (PDA), a cellular telephone, a smartphone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions **924** (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute instructions **924** to perform any one or more of the methodologies discussed herein.

(75) The example computer system **900** includes a processor **902** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a digital signal processor (DSP), one or more application-specific integrated circuits (ASICs), one or more radio-frequency integrated circuits (RFICs), or any combination of these), a main memory **904**, and a static memory **906**, which are configured to communicate with each other via a bus **908**. The computer system **900** may further include a graphics display unit **910** (e.g., a plasma display panel (PDP), a liquid crystal display

(LCD), a projector, or a cathode ray tube (CRT)). The computer system **900** may also include alphanumeric input device **912** (e.g., a keyboard), a cursor control device **914** (e.g., a mouse, a trackball, a joystick, a motion sensor, or other pointing instrument), a data store **916**, a signal generation device **918** (e.g., a speaker), and a network interface device **920**, which also is configured to communicate via the bus **908**.

(76) The data store **916** includes a machine-readable medium **922** on which is stored instructions **924** (e.g., software) embodying any one or more of the methodologies or functions described herein. The instructions **924** (e.g., software) may also reside, completely or at least partially, within the main memory **904** or within the processor **902** (e.g., within a processor's cache memory) during execution thereof by the computer system **900**, the main memory **904** and the processor **902** also constituting machine-readable media. The instructions **924** (e.g., software) may be transmitted or received over a network **926** via network interface **920**.

(77) While machine-readable medium **922** is shown in an example embodiment to be a single medium, the term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions (e.g., instructions **924**). The term “machine-readable medium” shall also be taken to include any medium that is capable of storing instructions (e.g., instructions **924**) for execution by the machine and that cause the machine to perform any one or more of the methodologies disclosed herein. The term “machine-readable medium” includes, but should not be limited to, data repositories in the form of solid-state memories, optical media, and magnetic media.

(78) In this description, the term “engine” refers to computational logic for providing the specified functionality. An engine can be implemented in hardware, firmware, and/or software. Where the engines described herein are implemented as software, the engine can be implemented as a standalone program, but can also be implemented through other means, for example as part of a larger program, as any number of separate programs, or as one or more statically or dynamically linked libraries. It will be understood that the named engines described herein represent one embodiment, and other embodiments may include other engines. In addition, other embodiments may lack engines described herein and/or distribute the described functionality among the engines in a different manner. Additionally, the functionalities attributed to more than one engine can be incorporated into a single engine. In an embodiment where the engines as implemented by software, they are stored on a computer readable persistent storage device (e.g., hard disk), loaded into the memory, and executed by one or more processors as described above in connection with FIG. 9. Alternatively, hardware or software engines may be stored elsewhere within a computing system.

(79) As referenced herein, a computer or computing system includes hardware elements used for the operations described here regardless of specific reference in FIG. 9 to such elements, including, for example, one or more processors, high-speed memory, hard disk storage and backup, network interfaces and protocols, input devices for data entry, and output devices for display, printing, or other presentations of data. Numerous variations from the system architecture specified herein are possible. The entities of such systems and their respective functionalities can be combined or redistributed.

## Claims

1. A system comprising: at least one validation node included in a decentralized peer-to-peer (P2P) computer network comprising at least one processor and memory storing instructions that cause the at least one validation node to perform: obtaining a media item to be included in a dataset for training an artificial intelligence (AI) model; applying pre-defined validation criteria to the media item, wherein the pre-defined validation criteria comprises one or more operations for modifying or evaluating the media item to establish validity; generating at least one data record associated with

the media item based at least in part on application of the pre-defined validation criteria, wherein the at least one data record indicates whether the media item is determined to be valid or invalid based on the application of the pre-defined validation criteria; and recording the at least one data record associated with the media item in a blockchain associated with the decentralized P2P computer network; and at least one recordation node included in the decentralized P2P computer network comprising at least one processor and memory storing instructions that cause the at least one recordation node to perform: determining a consensus on a validity of the media item based at least in part on a plurality of data records associated with the media item in the blockchain, the plurality of data records including the at least one data record generated by the at least one validation node; and recording, to the blockchain, an aggregate data record that includes information indicating the consensus determined for the validity of the media item, wherein the information provides at least an indication as to whether the media item satisfies the pre-defined validation criteria and is to be included in the dataset for training the AI model.

2. The system of claim 1, wherein said applying pre-defined validation criteria to the media item further causes the at least one validation node to perform: applying pre-defined digital image processing operations on the media item; and applying pre-defined operations for testing adversarial vulnerabilities in the media item.

3. The system of claim 2, wherein said applying pre-defined operations for testing adversarial vulnerabilities in the media item further causes the at least one validation node to perform: processing the media item based at least in part on one or more artificial intelligence (AI) models that are trained to evaluate the media item for adversarial vulnerabilities.

4. The system of claim 2, wherein the instructions further cause the at least one validation node to perform: detecting at least one adversarial vulnerability in the media item based at least in part on the pre-defined operations for testing adversarial vulnerabilities; and applying one or more operations to correct the at least one adversarial vulnerability detected in the media item.

5. The system of claim 1, wherein said generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria further causes the at least one validation node to perform: determining information describing a lineage associated with the media item, wherein the lineage identifies at least one training dataset in which the media item is included; and storing the information describing the lineage associated with the media item in the at least one data record.

6. The system of claim 1, wherein said generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria further causes the at least one validation node to perform: determining metadata information associated with the media item, wherein the metadata information includes at least one of image metadata associated with the media item, descriptive metadata associated with the media item, or administrative metadata associated with the media item; and storing the metadata information associated with the media item in the at least one data record.

7. The system of claim 1, wherein said generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria further causes the at least one validation node to perform: determining a hash value for the media item based on a digital cryptographic hash function; and storing the hash value for the media item in the at least one data record.

8. The system of claim 1, wherein said generating at least one data record associated with the media item based at least in part on the application of the pre-defined validation criteria further causes the at least one validation node to perform: determining one or more annotations associated with the media item; and storing the one or more annotations associated with the media item in the at least one data record.

9. The system of claim 1, wherein said determining a consensus on a validity of the media item based at least in part on a plurality of data records associated with the media item in the blockchain

further causes the at least one recordation node to perform: evaluating each of the plurality of data records to determine a respective validity determination associated with each data record; and determining the consensus on the validity of the media item based at least in part on a plurality of validity determinations determined for the plurality of data records.

10. The system of claim 1, said applying pre-defined operations for testing adversarial vulnerabilities in the media item further causes the at least one validation node to perform: reorienting or resizing the media item before the pre-defined validation criteria are applied to the media item.

11. A system comprising: at least one validation node that is included in a decentralized peer-to-peer (P2P) computer network and that comprises at least one processor and memory storing instructions that cause the at least one validation node to: obtain a media item to be included in a dataset for training an artificial intelligence (AI) model; determine whether the media item is valid by applying, to the media item, a pre-defined validation criterion that is representative of an operation for modifying or evaluating the media item; generate a data record for the media item that indicates whether the media item is determined to be valid or invalid based on an outcome of the operation; and record the data record generated for the media item in a blockchain that is associated with the decentralized P2P computer network; and at least one recordation node that is included in the decentralized P2P computer network and that comprises at least one processor and memory storing instructions that cause the at least one recordation node to: determine a consensus on a validity of the media item based on an analysis of a plurality of data records generated for the media item and recorded to the blockchain by a plurality of validation nodes, including the at least one validation node; and record, to the blockchain, a second data record that includes information describing the consensus on the validity of the media item, so as to indicate whether the media item satisfies the pre-defined validation criterion and is to be included in the dataset for training the AI model.

12. The system of claim 11, wherein the instructions further cause the at least one validation node to: determine information for the media item that is to be included in the data record.

13. The system of claim 12, wherein the information includes a hash value of the media item, details related to a provenance or a lineage of the media item, metadata that is associated with the media item, or any combination thereof.

14. The system of claim 11, wherein the instructions further cause the at least one validation node to: generate, for the media item, a hash value using a digital cryptographic hash function; and store the hash value generated for the media item in the data record.

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